



Case Report

Forensic identification of rape and murder case based on morphological features of hair: A case report

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ABSTRACT

In forensic identification, there are many cases of hair poisoning detection and analysis to assist forensic identification, with fewer reported cases using morphological features to assist forensic identification. Herein, we report the case of a 13-year-old girl who had been raped and killed. The body was abandoned in a snowy wilderness and was discovered 5 months later. After an on-site investigation and forensic examination, the case conformed to the characteristics of a homicide, and a strand of hair left on the deceased's scarf became key evidence. Through morphological methods such as stereomicroscopy, optical microscopy, and scanning electron microscopy, we believe that the hair conformed to the morphological and histological characteristics of pubic hair, which helped clarify the case. Based on this case, we believe that the morphological characteristics of hair are of great significance for forensic identification and that rich morphological observation methods can enhance the evidential value of hair in forensic cases.

1. Introduction

Microscopy and chemical analysis techniques have been used to analyze hair in forensic identification [1]. Morphological analysis of hair can be categorized as macroscopic and microscopic. Macroscopic analysis can be used to observe hair length, color, degree of curl, and tensile strength, which assist in making preliminary judgments regarding the source and type of hair [1]. In contrast, microscopic analysis offers a more detailed examination and can reveal internal structural characteristics of the hair, including pigment content, medullary index, elliptical index, and hair cuticle structure [2].

In this report, we present the case of a 13-year-old girl who was raped and murdered, with a strand of hair left on her scarf, serving as crucial evidence. Utilizing morphological techniques such as stereomicroscopy, optical microscopy, and electron microscopy, we believe that the hair exhibits the morphological and histological characteristics of pubic hair. While DNA testing of the hair follicle has successfully identified the suspect, our morphological observations of the hair continue to offer

significant leads for the investigation.

2. Case report

A 13-year-old girl went missing after playing outside. Five months later, her body was found in a snowy wilderness. A curved hair was extracted from the pink scarf of the deceased and sent to the Physical Evidence Appraisal Center of the Ministry of Public Security for inspection. DNA was detected in hair follicles, which matched the Y-chromosomal DNA of a local family. Through further investigation and sampling of that family, the suspect was ultimately identified. The STR typing of the hair found on the submitted pink scarf matched the genotypes at 17 STR loci of the suspect. To further clarify this case, the local public security organs sent the hair on the pink scarf, along with the suspect's head hair, pubic hair, and armpit hair, to our unit for morphological observation to determine the hair type.

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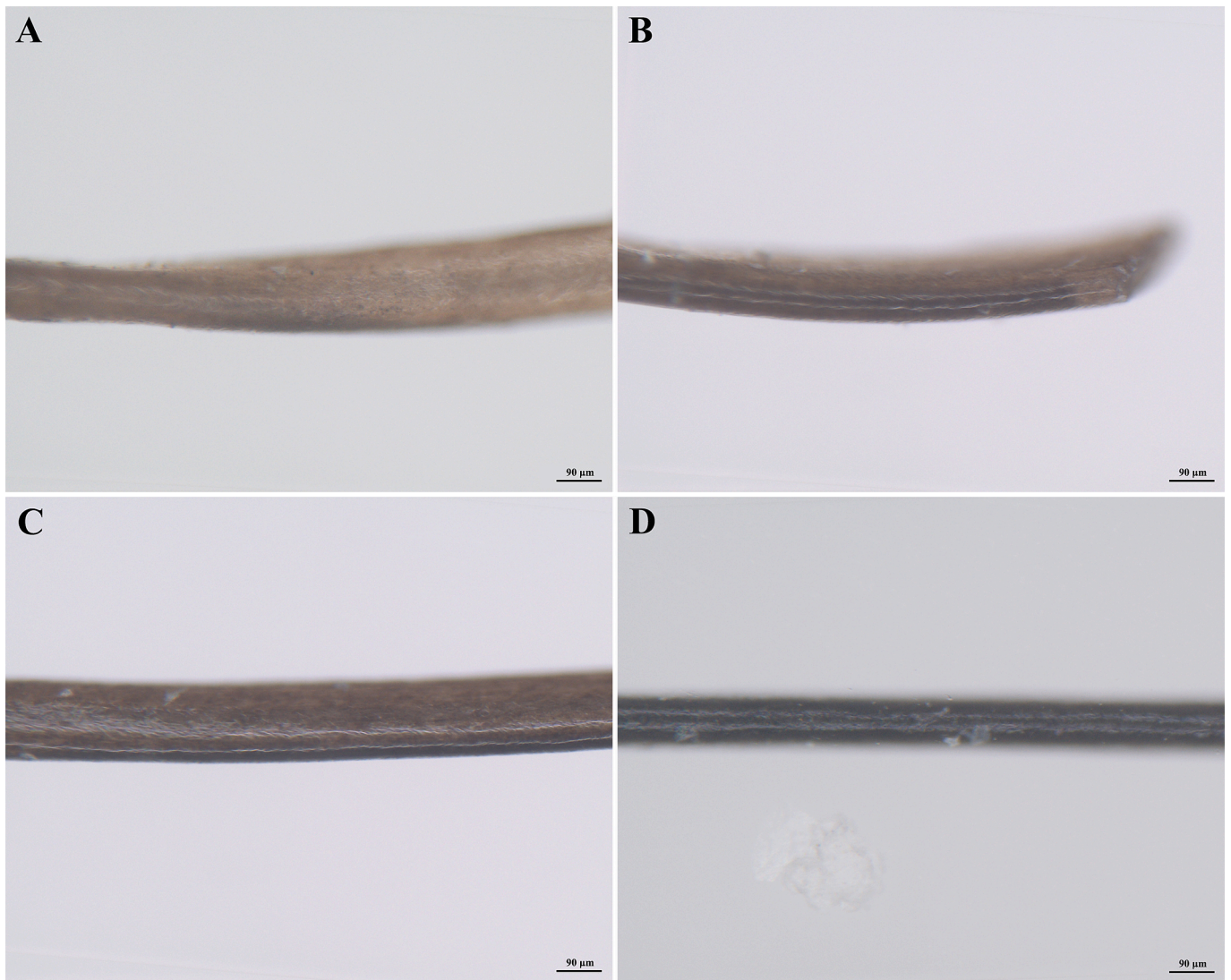


Fig. 1. Stereoscopic microscope examination showing (A) hair to be tested (100 $\times$), (B) suspect's pubic hair (100 $\times$), (C) suspect's armpit hair (100 $\times$), and (D) suspect's head hair (100 $\times$).

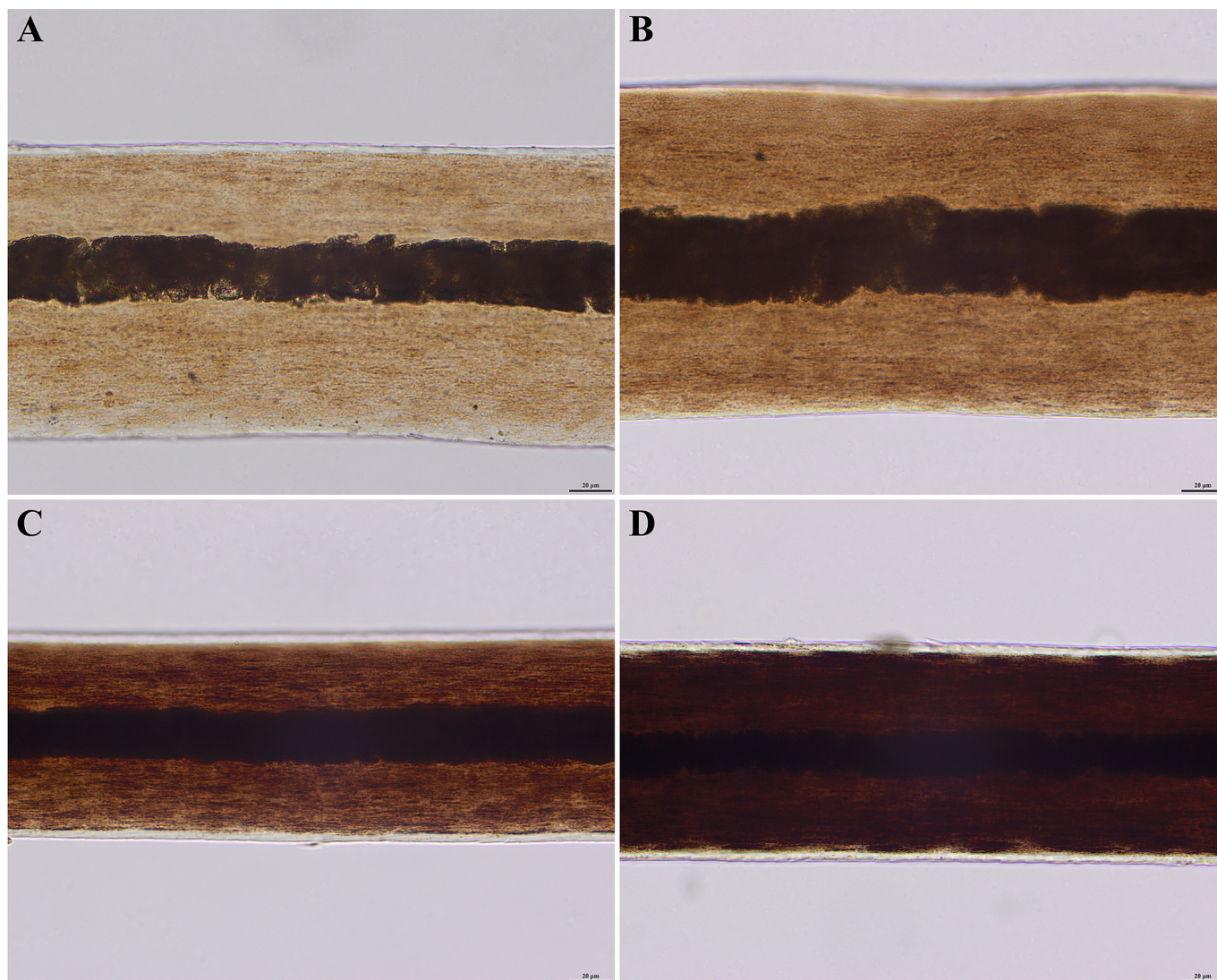


Fig. 2. Optical microscope examination showing (A) the hair to be tested (40 $\times$), (B) suspect's pubic hair (40 $\times$), (C) suspect's armpit hair (40 $\times$), and (D) suspect's head hair (40 $\times$).

2.1. Samples

The follicle of the hair to be tested was extracted for DNA testing. The suspect's hair samples sent for inspection included 8 pubic hairs, 16 armpit hairs, and one small bag of hairs. The hair was subjected to histological examination using stereomicroscopy, optical microscopy, and scanning electron microscopy. Since the hair provided from the crime scene for testing was from the middle section of the hair shaft, we selected the same region for observation in the other samples as well.

2.2. Stereomicroscopy

The hair shafts from all samples were cut and placed separately into clean petri dishes for direct observation under a stereomicroscope. The observation results are shown in Fig. 1. The hair to be tested (Fig. 1A) was 4.0 cm long and irregularly curled, with twisting visible in the curled areas. The suspect's pubic hairs (Fig. 1B) ranged from 4.0 cm to 6.0 cm in length and were irregularly curled. The suspect's armpit hairs (Fig. 1C) ranged from 4.0 cm to 6.0 cm in length and were slightly curly. The suspect's head hairs (Fig. 1D) were short and straight in shape, ranging from 0.1 to 0.5 cm in length.

2.3. Optical microscopy

The middle section of the hair shafts from all samples was cut, and after defatting, vacuuming, dehydrating, and sealing the hair samples, optical microscopy was performed (Fig. 2) [3]. The hair to be tested (Fig. 2A) was irregularly curled, with some curled areas showing hair torsion and low melanin content. The suspect's pubic hair (Fig. 2B) was irregularly curled and had low melanin content. The suspect's armpit hair (Fig. 2C) was slightly curly, with moderate melanin content between his pubic hair and hair. The suspect's head hair (Fig. 2D) was short and straight, with a higher content of melanin than the pubic and armpit hair, as well as the hair under test.

2.4. Electron microscopy

The middle section of the hair shafts from all samples was cut to prepare three longitudinal samples and three cross-sectional samples. Using the Czech TESCAN MIRA LMS scanning electron microscope, the longitudinal sections of the hair were observed at an energy of 2.00 keV, with the Det set to SE and a magnification of 3000 $\times$. For the cross-sectional observations, the energy was set to 5.00 keV, and the Det was set to SE and a magnification of 500 $\times$. The major axis, minor axis, and medullary length in the cross-sectional images were measured ($n =$

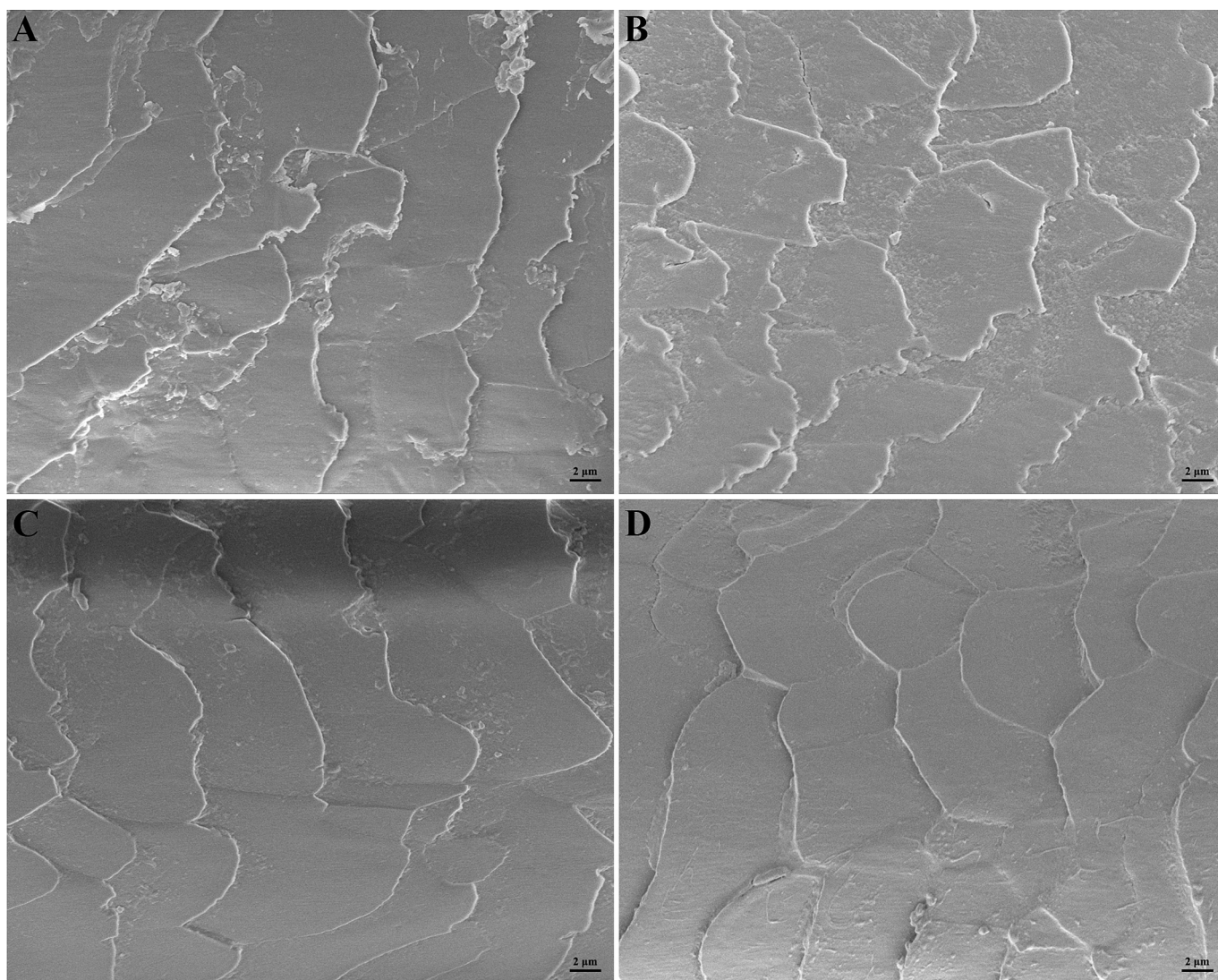


Fig. 3. Scanning electron microscopy examination showing (A) longitudinal profile of the hair to be tested (3000 $\times$), (B) longitudinal profile of the suspect's pubic hair (3000 $\times$), (C) longitudinal profile of the suspect's armpit hair (3000 $\times$), and (D) longitudinal profile of the suspect's head hair (3000 $\times$).

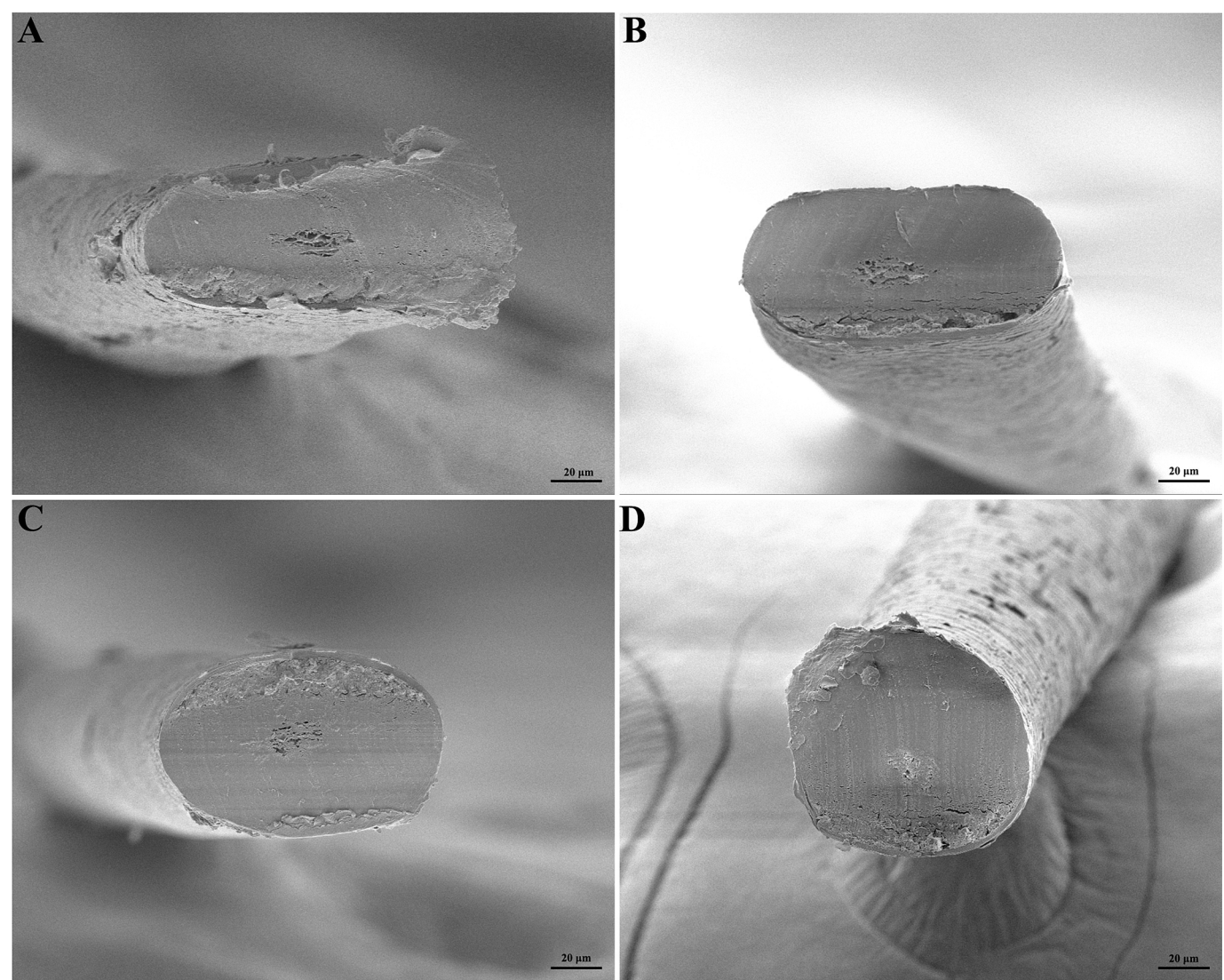


Fig. 4. Scanning electron microscopy examination showing (A) cross-section of hair to be tested (500×), (B) cross-section of suspect's pubic hair (500×), (C) cross-section of suspect's armpit hair (500×), and (D) cross-section of suspect's hair (500×).

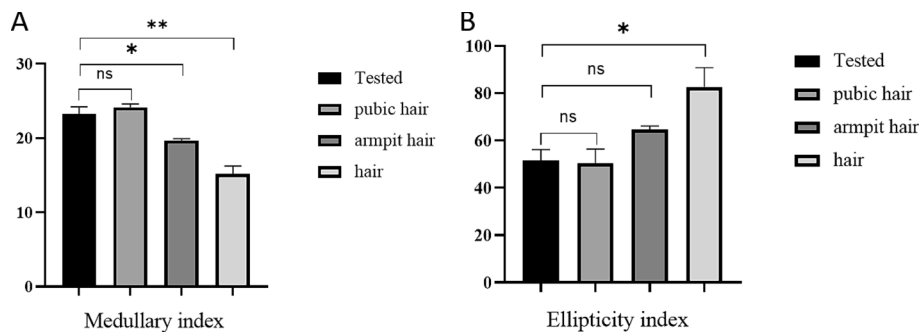


Fig. 5. The medullary index (A) and ellipticity index (B) of tested hair. (n = 3, *p < 0.05, **p < 0.01).

3). The results are shown in [Figs. 3–5](#) and [Supplementary Table](#). The cuticle of the hair to be tested ([Figs. 3A and 4A](#)) was arranged in a cross-shaped branching pattern with a jagged texture and irregular structure. The cross-section was elliptical, with an elliptical index of $51.59 \% \pm 4.62 \%$ and an average medullary index of $23.26 \% \pm 0.94 \%$. The cuticle of the suspect's pubic hair ([Figs. 3B and 4B](#)) was arranged in a cross-shaped branching pattern with a jagged texture and irregular

structure. The cross-section was elliptical, with an average elliptical index of $50.48 \% \pm 6.00 \%$ and an average medullary index of $24.14 \% \pm 0.47 \%$. The cuticle of the suspect's armpit hair ([Figs. 3C and 4C](#)) was arranged in a parallel pattern with a linear and smooth texture and regular structure. The cross-section was oval-shaped, with an average elliptical index of $64.63 \% \pm 1.47 \%$ and an average medullary index of $19.69 \% \pm 0.23 \%$. The cuticle of the suspect's head hair ([Figs. 3D and](#)

4D) was wavy in shape and had a regular structure. The cross-section was circular, with an average elliptical index of $82.75 \% \pm 8.03 \%$ and an average medullary index of $15.22 \% \pm 1.03 \%$.

2.5. Examination results

Based on the examinations conducted using the stereomicroscope, optical microscope, and scanning electron microscope, a comprehensive comparison was made between the tested hair, the suspect's pubic hair, armpit hair, and head hair. The analysis focused on several factors, including the degree of curvature, melanin content, ellipticity index [3], medullary index [4], and hair cuticle structure. The findings indicate that the hair found at the crime scene aligns more closely with the morphological and histological characteristics of the suspect's pubic hair. In the subsequent case trial, the suspect confessed to coercing the victim into engaging in oral sex, and our appraisal results played an important auxiliary role in clarifying the case.

3. Discussion

Hair characteristics vary between different races and between individuals [2,4,5]. Therefore, individual identification or population differentiation through the morphological observation of hair is unreliable. However, hair morphology can be beneficial for identifying hair from different parts of the same individual and may play an auxiliary role in forensic cases. Forensic experts should take this benefit into consideration.

We observed the degree of curvature, melanin content, hair cuticle structure, and indicators such as the medullary index and ellipticity index. In terms of melanin content, Asians usually have dark hair, and their melanin content is higher compared to other ethnic groups. In terms of medullary index, African hair is more likely to have a continuous or discontinuous medulla, with a medullary ratio of about 0.20–0.25 [5], while the medullary ratio of the Japanese population ranges from approximately 0.17–0.25 [4]. In terms of the ellipticity index, African hair is mostly horizontally elliptical, European hair is mostly semi-elliptical, and Asian hair is mostly circular [6]. These characteristics may overlap among the different ethnic groups. The hair cuticle structure is often used to distinguish between human and animal hair [7]. There may be differences in the number of layers, width, and inclination angle of the cuticle between different ethnic groups [8]. In this case, the suspect's hair matched the characteristics typical of the Asian population. The morphological observation results of hair should not rely on any single feature for evaluation, instead requiring comprehensive analysis of multiple indicators while considering racial differences and population diversity. In contrast, the diameter of the hair can vary along the entire length of the hair shaft, with the axial diameter near the root of the hair possibly thicker and thinner at the distal end due to natural wear or damage. Therefore, multiple measurements must be performed at locations where hair shaft thickness is uniform. The daily state of the hair from various sections also differs. The ends of pubic and armpit hair are mostly naturally grown and worn, whereas the ends of head hair are mostly cut sections that need to be carefully identified.

At the same time, this case has certain limitations. The hair recovered from the crime scene had the portion containing the hair follicle removed for DNA testing, resulting in a limited length available for our analysis. Consequently, there is a possibility of false positives in our findings, which cannot be considered direct evidence. Nonetheless, the observation of hair morphology warrants the attention of forensic experts, as the various morphological methods may serve as a valuable complement to forensic science.

In summary, this case used the morphological observations of hair and various hair characteristic indicators to distinguish hair from different parts of the same individual to clarify the case and provide new insights for forensic hair morphology research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.legalmed.2025.102613>.

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